

Sign Glove

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I. INTRODUCTION

The group proposes a project about sign to speech gloves, using gyroscope sensors to detect and track movement and signs and convert them to speech using AI.

There are several commercially available and experimental projects demonstrating functional sign-to-speech gloves.

One notable example is BrightSign, which emphasizes a wide library of supported languages, customizable voice options, and a companion app that enhances the glove's usability [1].

Another significant development was presented by UCLA researchers in 2019, who created a prototype capable of recognizing American Sign Language (ASL) gestures, aiming for real-time translation into text and speech [2].

While these projects demonstrate the feasibility of translating sign language through wearable technology, they often focus on predefined gesture libraries, relatively high costs, or require specialized hardware and software ecosystems.

Our project seeks to explore alternative approaches to make sign-to-speech translation more accessible, adaptable, and practical. This by using simple sensor technology and open source tools to achieve hand gesture recognition within the constraints of a rapid development cycle.

A. Problem Formulation

- Are gyroscope sensor suitable sensors for use in a sign glove? While being a cheaper option, will it provide accuracy and work reliably in real-time?
- What are the practical limitations and possibilities with building a low cost sign glove with limited resource under a short development cycle?

II. METHOD

The idea is to use gyroscopes on each finger in order to acquire a digital model of the fingers positions which can then be used to recognize signs in sign-language. These sensors will be connected to a microcontroller (e.g., Arduino) mounted on the glove along with a portable power supply. The data collected will be used to recognize static and dynamic signs via a neural network, allowing for speech output.

Planned Timeline (3–4 weeks, 3 people, 20 hours/week):

- Week 1: Planning, research, and choice of sensors.

- Week 2: Prototype, basic code.
- Week 3: Signs and Fine-tuning implementation.
- Week 4: AI recognition (if there is time).

For components, one gyroscope sensor per finger, an Arduino to use as the brain, and a small power supply will be used.

Some risks that could occur include sensor noise that results in unreliable gesture recognition. Hardware integration proves to be more difficult to implement than predicted and therefore take up more time than anticipated. And with a time constraint it limits the depth of testing and tuning that can be done. Outside of the 3-week plan, if there would be time for AI integration the machine learning model could prove difficult.

III. EXPECTED RESULTS

We expect to develop a working prototype capable of creating a real-time digital model of the hand using gyroscope data. The glove should be able to discriminate between signs. And if time allows for it, the group aims to implement a machine learning model to improve the sign recognition.

REFERENCES

- [1] Centers for Disease Control and Prevention (CDC), "Brightsignglove," <https://www.brightsignglove.com/>, accessed 29-04-2025.
- [2] M. Chin, "Wearable-tech glove translates sign language into speech in real time," <https://newsroom.ucla.edu/releases/glove-translates-sign-language-to-speech>, accessed 29-06-2019.